

ECON 101

TA Worksheet, Module 13 (Profit and Pricing)

Name: _____

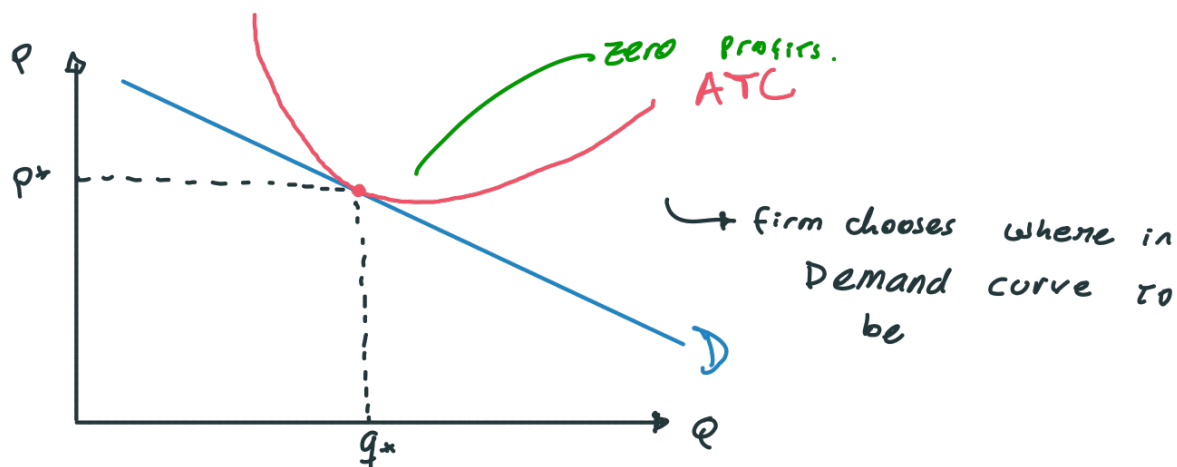
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- Let's suppose Elon Musk somehow manages to make Twitter "profitable" and earns \$200 million in accounting profit. Suppose the \$44 billion he used to buy Twitter could have bought the entire NBA. And suppose he would have earned \$600 million in accounting profit from the NBA. What is Musk's economic profit from Twitter?

Economic profit is considering opportunity costs.

Suppose \$44 B in NBA was "next" best thing
- 400 million is the economic profit. (200 - 600)

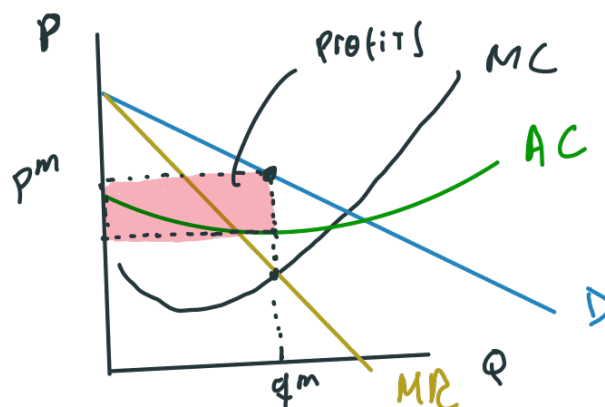
- Draw a picture of an imperfectly competitive firm in the long run in a market with unrestricted entry. Label everything! (hint: think about profit and the average cost curve).



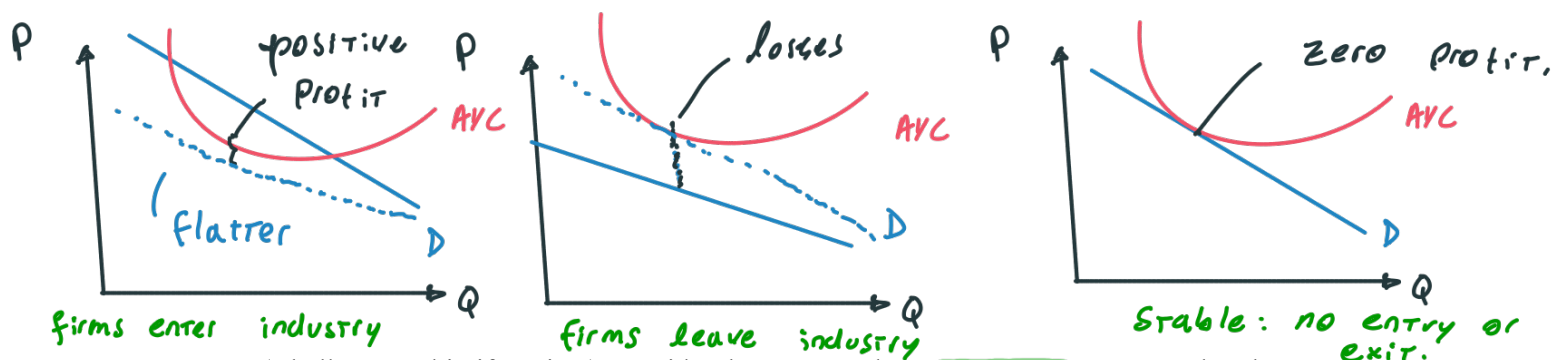
- What info do you need to figure out an imperfectly competitive firm's quantity, price, and profit?

Need to set $MC = MR$, get Q , plug in D and deduct costs. So, need:

- MC
- ATC
- MR
- D



4. Assuming unrestricted entry, describe (with words and pictures) how an industry moves to zero economic profits.



5. (Challenge – skip if no time) Consider the taco stands on State Street. Suppose that the taco market on State Street is a perfectly competitive market, where all taco stands are exactly the same (i.e. producing the exact same tacos and having the exact same cost features). The market demand for tacos is given by the demand curve where P is the price per taco and Q is the market quantity of tacos:

$$P = 100 - 2Q$$

Each taco stand faces a marginal cost curve given by the following equation where q is the quantity of tacos produced by the firm:

$$MC = 4q$$

and a total cost curve given by:

$$TC = 2q^2 + 8$$

- a) For each taco stand, what is the fixed cost (FC)? What is the average fixed cost (AFC) curve and the average total cost (ATC) curve?

TC is $2q^2 + 8$. 8 is fixed cost. AFC is $\frac{8}{q}$
 ATC is $2q + \frac{8}{q}$

- b) What is the break-even price for each taco stand (when profit = 0)? Hint: it's where $ATC = MC$. Warning: the quantity will seem really low.

$$4q = 2q + \frac{8}{q} \Leftrightarrow 2q = \frac{8}{q} \Leftrightarrow q^2 = 4 \Rightarrow q = 2$$

$$P = MC \Leftrightarrow P = 8$$

- c) In the long run, what would the price for one taco be on State Street? How many taco stands will stay in the market?

$\rightarrow P = ATC$. Plug $P = 8$ into $P = 100 - 2Q$.
 We have $8 = 100 - 2Q \Leftrightarrow Q = 46$. Each firm produces 2, so 23 firms.

- d) Assume that the current price for one taco on State Street is \$12. How many tacos does each taco stand produce? Suppose that there is no cost in setting up/closing down a taco stand. What will happen to the number of taco stands?

$$12 = 4q \Leftrightarrow q = 3. \text{ There are } 12 = 100 - 2Q \Rightarrow Q = 44$$

There are ~ 14.667 stands. Long-run should be 23. There will be entry.

6. What does price elasticity of demand have to do with price discrimination?

Seller could charge more to inelastic consumers and less to elastic ones.

7. Describe a time that you paid more or less due to price discrimination (try to avoid the student discount example). Was it 3rd degree (group pricing) or second degree (hurdle) price discrimination?

Paying less in plane tickets when you search more, look prices more frequently and start looking earlier. (Hurdle).

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TA Worksheet, Module 14 (Game Theory)

- In the 1960s, when people were dumb, there was a game called “chicken” that went like this: two people drive their cars at each other. Before they crash, the drivers have to choose whether to keep going straight or turn. If both go straight, they crash. But if one goes straight and the other turns, then the one who goes straight is super cool and the other player sucks. If they both turn then it’s just kinda lame.

- Here is the payoff matrix. What is/are the equilibria to this game? (assume it’s better to lose than crash and better to tie than lose.)

	Player 2 Straight	Player 2 Turn
Player 1 Straight	Crash, Crash	1 wins, 2 loses
Player 1 Turn	1 loses, 2 wins	Tie, tie

Equilibria: - P1 Turn, P2 straight
- P1 straight, P2 turn.

- What kind of game is this? Anti-coordination

- Suppose both players played the maximin strategy in the chicken game. What would be the result?

Both would avoid crashing. Strategy is [Turn, Turn]
The outcome would be [Tie, Tie]

- This game has no story, it’s just a game. What’s the Nash Equilibrium of this game? (the first number is player 1’s payoff, the second number is player 2’s payoff.)

	Player 2 Left	Player 2 Right
Player 1 Top	300, 200	100, 300
Player 1 Bottom	200, 100	200, 200

The Nash Equilibrium is P1 Bottom, P2 Right.
The outcome is [200, 200].

3. If these firms could collude (through a repeated game), what choices would they make to maximize their joint profits? (they are choosing the quality of their product)

	Firm 2 chooses high quality	Firm 2 chooses low quality
Firm 1 chooses high quality	Firm 1 profits = \$2 million, Firm 2 profits = \$2 million	Firm 1 profits = \$5 million, Firm 2 profits = \$1 million
Firm 1 chooses low quality	Firm 1 profits = \$1 million, Firm 2 profits = \$5 million	Firm 1 profits = \$4 million, Firm 2 profits = \$4 million

Colluding means maximizing joint profits. Both choose Low

	High	Low
Low	4	6
High	6	8

4. In a repeated game, what's the difference between the grim trigger strategy and the tit-for-tat strategy?

Grim Trigger: Punish forever

Tit-for-tat: Copy what other player does

↳ "if you play nice, I play nice".